REASONING • RAS PRELIMS NOTES

Reasoning & Mental Ability

Methods, formulas and worked examples - statement reasoning, series, coding, arrangements and complete basic numeracy

Inside these notes

- Every topic taught as a method first, then solved examples
- Full formula bank - percentage, interest, area, mean, probability
- A trap callout at the end of every chapter

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Notes made for RAS aspirants - simple language, connected concepts, PYQ-based facts.

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Statement-Based Reasoning

This chapter covers the five statement types RAS asks again and again. In every type, one rule sits above all others: **use only what is inside the statement**. Do not use your own knowledge, your own opinion, or what is "generally true" in the outside world. The examiner's world begins and ends with the printed lines. Read the statement as if it is 100 percent true, then test each option against the rules given below.

Statement and Assumptions

An **assumption** is something taken for granted — a hidden belief on which the statement stands. The question gives a statement and two assumptions. You must decide which assumption is **implicit**, meaning the speaker must have believed it, otherwise the statement would make no sense.

Rules to apply:

- An assumption must be **directly connected** to the statement. If it brings in a new topic, reject it.
- An assumption is implicit if the statement **falls apart** without it. Ask: "If this were false, would the statement still be said?" If not, it is implicit.
- Words like **only, all, always, never** make an assumption too strong. Such assumptions are usually NOT implicit.
- A **suggestion, request or advertisement** assumes the listener can act on it and that the suggested thing is possible and useful.
- An assumption about a **past cause** is implicit only when the statement clearly hints at that cause.
- An assumption that repeats the statement in different words is NOT implicit — it is the statement itself.

Test	Question to ask yourself
Necessity test	Does the statement collapse if this is false?
Connection test	Is it about the same subject as the statement?
Strength test	Does it use extreme words like only/all/never?
Possibility test	Does the statement treat this as possible?

- **Example 1 — Statement: "Take the early morning train to reach Jaipur in time for the interview." Assumptions: I. Trains run in the early morning on this route. II. The interview is in Jaipur.**

- Step: Advice to take a morning train is useless unless such trains exist, so I is implicit. The statement itself says the interview is reached by going to Jaipur — II only repeats the statement, so II is not an assumption. - Answer: **Only I is implicit.**

- **Example 2 — Statement: an advertisement says "Use herbal soap for glowing skin." Assumption: people want glowing skin.**

- Step: An advertisement is printed only because the maker assumes people desire the promised result. Without that desire, the ad has no purpose. - Answer: **The assumption is implicit.**

MEMORY HOOK: An assumption is the silent leg the statement stands on — remove the leg, and if the statement falls, the assumption was implicit.

EXAM TRAP: The classic mistake is marking an assumption implicit just because it is TRUE in real life. Truth is not the test — necessity to the statement is the test.

PYQ PATTERN: RAS loves advertisement and notice-board statements; in notices, "the notice will be read and followed" is almost always implicit.

Statement and Arguments

Here a decision or policy is given, followed by two arguments — one in favour (say "yes, should be done") and one against. You must judge which argument is **strong**. A strong argument is directly related to the issue, important, and based on logic or established fact. A weak argument is emotional, vague, based on example alone, or attacks something beside the point.

- **Strong**: touches the core of the issue; speaks of benefit/harm to society, economy, health, law; uses words like "it will reduce unemployment".
- **Weak**: uses **only, alone, merely**; copies another country without reason; rests on emotion ("it is our culture, how dare we"); compares with an unrelated thing; is a simple yes/no restatement.
- An argument that says "this is the **only way**" is weak unless the statement proves no other way exists.
- An argument based on a **famous person or a foreign example** alone is weak.
- **Example — Statement: Should plastic bags be banned? Argument I: Yes, plastic chokes drains, kills cattle and stays in the soil for hundreds of years. Argument II: No, shopkeepers find plastic bags very convenient.**

- Step: I attacks the core harm with real, large-scale effects — strong. II offers minor convenience against public harm — weak. - Answer: **Only Argument I is strong.**

EXAM TRAP: Students mark the argument they personally agree with as strong. Never judge by your opinion — judge by the argument's quality: relevance, importance, logic.

Statements and Conclusion

A **conclusion** is a fact that can be **drawn directly** from the given statements. The statement may be one line or many. Whatever the statements say is true — even if it sounds odd ("all pens are trees"). Your job is only to check whether the conclusion follows with certainty.

- The conclusion must follow in **every possible case**, not just in some cases. If it is only "possible", it does not follow (unless the question asks for possibility).
- Do not bring **outside facts**. If the statement says all cats are dogs, then in this question, cats are dogs.
- A conclusion that introduces a **new word or idea** not in the statement does not follow.
- **Complementary pairs**: when neither conclusion alone is certain but they form an either-or pair (one must be true), the answer is "either follows". This needs the two conclusions to cover all possibilities between the same two terms.
- **Example — Statements: All books are pens. All pens are chairs. Conclusions: I. All books are chairs. II. Some chairs are books.**

- Step: Books sit inside pens, pens sit inside chairs, so books are inside chairs — I follows. Part of chairs is books, so II also follows. - Answer: **Both conclusions follow.**

EXAM TRAP: The biggest trap is choosing a conclusion that is "generally true in life" but not proved by the statements. Life stays outside the exam hall.

Statement and Courses of Action

A **course of action** is a practical step that should be taken to solve a problem stated in the passage. You judge whether the suggested step logically follows.

A course of action FOLLOWS when:

- It is a **practical, workable step** that reduces or solves the stated problem.
- It is taken by the **right authority** (government, management, police) for that problem.
- It is **proportionate** — not an extreme reaction like "close down the whole department" for a small fault.

A course of action does NOT follow when:

- It is **extreme or harsh** (ban everything, dismiss everyone) for a limited problem.
- It creates a **new, bigger problem**.
- It is based on something **not mentioned** in the statement.
- It is a **long-term dream** ("poverty should be removed") instead of a step.
- **Example — Statement: Many students of a school fell ill after eating the mid-day meal. Courses of action: I. The food samples should be tested and the cook suspended till the report comes. II. The mid-day meal scheme should be stopped in all schools of the state forever.**

- Step: I is a direct, proportionate, workable step. II is extreme and punishes everyone for one incident. - Answer: **Only I follows.**

EXAM TRAP: Any course of action with "all", "forever" or "completely banned" is almost always rejected — examiners reward measured steps, not dramatic ones.

Analytical and Critical Reasoning

This is the mixed basket: cause-effect, assertion-reason, and small logic puzzles of everyday reasoning.

Cause and Effect

Two events are given. Decide if the first is the **cause** and the second its **effect**, or the reverse, or both are independent, or both are effects of a common cause.

- The cause must come **before** the effect in time and must be **capable** of producing it.
- If both events can be produced by one bigger event (drought causes both price rise and migration), mark **common cause**.

Assertion and Reason

Assertion (A) is a claim; Reason (R) tries to explain it. Four outcomes: both true and R explains A; both true but R does not explain A; A true R false; A false R true. First check each truth separately, then check the link — most errors happen at the link step.

- **Example — A: Ice floats on water. R: Ice is less dense than water.**

- Step: Both are true, and lower density is exactly why ice floats, so R correctly explains A. - Answer: **Both true, and R is the correct explanation of A.**

MEMORY HOOK: For cause-effect, always ask "which happened first, and could the first really produce the second?"

EXAM TRAP: In assertion-reason, students rush to the explanation link before checking simple truth — a false reason can never explain anything.

Syllogisms and Logical Venn Diagrams

A **syllogism** gives two or three statements using **all, some, no** and **only**, and asks which conclusions follow. The statements are always true in the question's world. The safest method is the Venn diagram method — draw circles and let the picture decide.

The Four Statement Types

Statement	Meaning	Venn picture
All A are B	Every A is inside B	Circle A fully inside circle B
No A is B	A and B share nothing	Two separate circles
Some A are B	At least one A is B	Circles overlap; cross (x) in overlap
Some A are not B	At least one A lies outside B	Cross in the part of A outside B

The Venn Method — Step by Step

- Step 1: Draw the **biggest (universal) relation first** — draw "all" and "no" relations before "some".
- Step 2: Put a **cross (x)** for every "some" — the cross shows a member definitely exists there.
- Step 3: Test each conclusion against the finished diagram. A conclusion follows only if it is true in **every diagram** that satisfies the statements.
- Step 4: If a conclusion is true in one drawing but false in another valid drawing, it does **not** follow.
- Step 5: For "some", remember it means **at least one** — it also allows the possibility of "all", unless "no" blocks it.

Standard Conclusions Table

Given	Definite conclusion	NOT definite
All A are B	Some B are A	All B are A
No A is B	No B is A; All A are not B	—
Some A are B	Some B are A	All A are B
Some A are not B	Some B may or may not be A	No A is B

"Only" and Possibility Cases

- **"Only A are B"** means **All B are A**. The word "only" reverses the direction — read it twice before drawing.
- **Possibility conclusions**: "All A being B is a possibility" follows when no statement blocks A from going fully inside B. If a "no" or "some not" blocks it, the possibility fails.
- A possibility conclusion fails when a **definite opposite** is already proved by the diagram.

- **Example 1 — Statements: All pens are books. Some books are chairs. Conclusions: I. Some pens are chairs. II. Some books are pens.**

- Step: Pens are inside books; the chair overlap touches only some part of books, which may or may not include pens — so I is not certain. Part of books is definitely pens, so II follows. - Answer: **Only II follows.**

- **Example 2 — Statements: No cat is a dog. Some dogs are cows. Conclusions: I. Some cows are not cats. II. Some cows are cats.**

- Step: The dogs that are cows cannot be cats (no cat is a dog), so those cows are definitely not cats — I follows. Whether other cows are cats is unknown — II does not follow. - Answer: **Only I follows.**

MEMORY HOOK: Draw "all" and "no" first, drop the cross for "some" last — and a conclusion must survive every possible drawing.

EXAM TRAP: "Some A are B" never proves "some A are not B". Students add this extra conclusion from real-life feeling; the diagram does not give it.

Number and Letter Sequences

A **series** is a list of numbers or letters following one hidden rule. Your job is to find the rule from the first few terms and apply it once more. Always work left to right and check the rule against **every** given term, not just the first two.

Number Series — Types Table

Type	How it looks	How to spot it
Constant difference	5, 9, 13, 17... (+4)	Differences all equal
Increasing difference	2, 3, 6, 11, 18... (+1,+2,+3,+4 pattern in gaps)	Gaps grow by a fixed amount
Squares / cubes added	2, 6, 12, 20, 30 (+4,+6,+8,+10) or +1, +4, +9, +16	Gaps are squares or even numbers
Prime numbers	2, 3, 5, 7, 11, 13...	Terms themselves are primes
Multiply-add pattern	3, 7, 15, 31 (x2+1)	Each term about double the last
Two operations alternating	2, 6, 4, 8, 6, 10 (+4, -2, +4, -2)	Rise-fall-rise-fall pattern
Interleaved series	2, 3, 4, 6, 6, 9, 8, 12	Two series mixed — split odd and even places

Method steps:

- Step 1: Look at the **size of growth**. Slow, steady growth means addition; fast growth means multiplication or squares/cubes.
- Step 2: Write the **differences** between terms. If differences repeat or grow in a pattern, you have found the rule.
- Step 3: If differences look random, try **multiply-add** (each term = previous x k + c) or split the series into **two interleaved series**.
- Step 4: Verify the rule on all given terms, then produce the next term.
- **Example 1 — 7, 14, 23, 34, 47, ?**

- Step: Differences are 7, 9, 11, 13 — odd numbers increasing by 2. Next difference is 15. - Answer: **47 + 15 = 62.**

- **Example 2 — 5, 11, 23, 47, 95, ?**

- Step: Each term is previous x 2 + 1: 5x2+1=11, 11x2+1=23, 23x2+1=47, 47x2+1=95. - Answer: **95 x 2 + 1 = 191.**

- **Example 3 — 3, 4, 6, 9, 13, 18, ?**

- Step: Differences are 1, 2, 3, 4, 5 — increasing naturally. Next difference is 6. - Answer: **18 + 6 = 24.**

Finding the Wrong Term

Some questions do not ask for the next term — they give a complete series with **one wrong term** hidden inside and ask you to find it. The method is the same difference method, with one extra habit: the wrong term usually breaks the pattern at **two** places, because it disturbs the difference before it and the difference after it. First guess the clean pattern from the majority of terms, rebuild the whole series from that pattern, and the term that refuses to match your rebuilt list is the wrong one. Never trust the first two terms alone — a wrong term sitting early in the series can poison a rule built only on them, so test your guessed rule against the last terms as well.

- **Example — Find the wrong term: 2, 5, 10, 17, 27, 37.**

- Step: Differences are 3, 5, 7, 10, 10 — the clean pattern should be odd differences 3, 5, 7, 9, 11, giving the series 2, 5, 10, 17, 26, 37. The fourth difference should be 9, not 10, and the fifth should return to 11 (37 - 26), not 10. - Answer: **27 is the wrong term; it should be 26.**

Letter Series — The Alphabet Chart

Letter series are number series in disguise. Convert letters to positions, solve, convert back.

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

MEMORY HOOK: EJOTY — E=5, J=10, O=15, T=20, Y=25. These five anchor points let you count any letter's position in seconds.

MEMORY HOOK: Opposite letters add to 27 — A-Z, B-Y, C-X, D-W, E-V, F-U, G-T, H-S, I-R, J-Q, K-P, L-O, M-N. If a letter is past M, subtract from 27 to get its pair.

- **Example — B, D, G, K, P, ?**

- Step: Positions 2, 4, 7, 11, 16 — differences 2, 3, 4, 5. Next difference 6, so position 22. - Answer: **V.**

EXAM TRAP: In letter series, after Z the alphabet **wraps around** to A (26 + 2 = B, not a letter beyond Z). Forgetting the wrap-around is the most common letter-series error.

PYQ PATTERN: RAS frequently mixes one number series and one letter series in the same paper — practise converting letters to numbers until it is automatic.

Coding-Decoding

In **coding-decoding**, a word is written in a hidden code, and you must crack the rule from the example and apply it to a new word. There are only a handful of rules — once you can name the type, the question is finished.

The Main Types

Type	Example pattern	How it works
Letter shifting	CAT becomes DBU	Every letter moves forward/back by a fixed number
Shifting with pattern	ACE becomes BDG... steps +1, +2, +3	The shift changes in a pattern
Opposite-letter coding	CAT becomes XZG	Each letter replaced by its opposite (sum 27)
Number coding	CAT becomes 3120	Letters replaced by position numbers, maybe added
Word substitution	"water is called air"	One word is renamed; answer with the new name
Mixed letter-number	C3A20 style	Letters and positions mixed in place

Method Steps

- Step 1: Write the **positions** of the example word's letters and the code's letters/numbers below them.
- Step 2: Compare position by position — find the shift (+ or -) or the pairing rule.
- Step 3: Check the rule on **every letter** of the example, including the last one.
- Step 4: Apply the same rule to the target word, one letter at a time. Never guess the whole answer at once.
- **Example 1 — If CAT is coded as DBU, how is DOG coded?**

- Step: C(3) to D(4) is +1, A(1) to B(2) is +1, T(20) to U(21) is +1. Apply +1: D to E, O to P, G to H. - Answer: **EPH.**

- **Example 2 — If RAM is coded as IZN, how is SUN coded? (opposite letters)**

- Step: R(18) pairs with I(9) since 18+9=27; A(1) with Z(26); M(13) with N(14). So use opposites: S to H, U to F, N to M. - Answer: **HFM.**

- **Example 3 — In a code, "cat is called dog, dog is called fish, fish is called bird". Which one swims in water?**

- Step: In real life a fish swims. But in this code, fish is called bird. The question asks in the code language. - Answer: **Bird.**

MEMORY HOOK: First write positions, then compare — coding questions are arithmetic wearing a costume.

EXAM TRAP: In substitution questions, students answer with the real-world word. The answer must be the **code name**, not the real name. Also, in shifting codes, always check the last letter — some codes shift +1 for odd places and +2 for even places.

Blood Relations and Direction Sense

Blood Relations — The Generation Method

Blood relation questions give a chain like "A is the mother of B, B is the brother of C". Do not hold the chain in your head — draw it. Place people in **generations**: grandparents on the top line, parents in the middle, children below. Use a square for male, a circle for female, a double line (=) between husband and wife, a vertical line for parent-child, and a horizontal line joining siblings.

Relation Chart — Words You Must Know

Relation	Meaning
Paternal uncle/aunt	Father's brother / father's sister
Maternal uncle/aunt	Mother's brother / mother's sister
Nephew / Niece	Brother's or sister's son / daughter
Cousin	Uncle's or aunt's child
Father-in-law / Mother-in-law	Spouse's father / mother
Brother-in-law / Sister-in-law	Spouse's sibling, or sibling's spouse
Grandson / Granddaughter	Son's or daughter's child
Only son/daughter of my father	The speaker himself/herself (if no siblings stated)

Method steps:

- Step 1: Start from the **last person** in the chain and build backwards — the final link usually connects to the person asked about.
- Step 2: Mark **gender** wherever the relation word fixes it (mother, brother, wife). Where gender is unknown, keep both options open.
- Step 3: Count generations, not names. Uncle/aunt = parent's generation sibling; cousin = your own generation.
- **Example 1 — Pointing to a photo, Ravi says, "She is the daughter of my grandfather's only son." Who is she to Ravi?**

- Step: Grandfather's only son is Ravi's father. The father's daughter is Ravi's sister. - Answer: **She is Ravi's sister.**

- **Example 2 — A is the brother of B. B is the mother of C. What is A to C?**

- Step: B is C's mother; A is B's brother, so A is C's mother's brother — maternal uncle. - Answer: **A is the maternal uncle of C.**

Coded Relations

Here symbols carry relations: for example, **A + B means A is the mother of B, A - B means A is the brother of B, A x B means A is the father of B, A / B means A is the sister of B**. The symbols change from question to question — always read the key given in that question, never carry one question's key into the next. Solve left to right, building the family tree link by link, and answer only the pair asked about.

Direction Sense — The Basics

Fix the map in your mind: **North up, South down, East right, West left**. Between them sit NE, SE, SW, NW. A turn question depends on the direction the person is **currently facing**: left of west is

south; right of north is east, and so on. The safe method is to turn your own mental compass at each step — write the facing direction after every single move.

MEMORY HOOK: Shadow rule — in the morning the sun is in the east, so every shadow points **west**; in the evening the sun is in the west, so every shadow points **east**. Use it like this: in the morning, if your shadow falls exactly on your right side, your right hand points west — and the right hand points west only when you face **south**.

Turns and Shortest Distance

- Turn **left** = 90 degrees anti-clockwise from current facing; turn **right** = 90 degrees clockwise. A "turn back" or "turns around" means 180 degrees.
- For the **shortest distance** between start and end, when the path makes a right-angled corner, use **Pythagoras**: (shortest distance)² = (one side)² + (other side)². Common triples: 3-4-5, 6-8-10, 5-12-13.
- Net the movements: cancel equal opposite moves (3 km north and 3 km south cancel out) before applying Pythagoras.
- **Example 1 — A man walks 3 km east, turns left and walks 4 km. How far is he from the start?**

- Step: Facing east, left is north, so he goes 3 east and 4 north. Distance = root of ($3^2 + 4^2$) = root of 25. - Answer: **5 km from the start, in the north-east direction.**

- **Example 2 — Meena faces west. She turns right, then right again, then left. Which way does she face now?**

- Step: Facing west, right gives north; facing north, right gives east; facing east, left gives north. - Answer: **She faces north.**

EXAM TRAP: In relation questions, "my father's only son" fools students into searching for another person — it is usually the speaker himself. In direction questions, the classic trap is applying left/right from YOUR facing instead of the walker's current facing after each turn.

Ranking and Sitting Arrangement

Ranking — The One Formula

In a row or queue, if a person's position is known from both ends:

Total persons = (position from left) + (position from right) - 1

We subtract 1 because the person is counted twice — once from each side. From this one line come all ranking answers:

- **Position from right = Total - position from left + 1**
- **Position from left = Total - position from right + 1**
- If two people swap seats, the **new position of the second = old position of the first + (change in the first's position)**. Or: after a swap, total = (one's new position from one end) + (other's old position from the other end) - 1.
- **Example 1 — In a row, Kiran is 14th from the left and 9th from the right. How many persons are in the row?**

- Step: Total = 14 + 9 - 1. - Answer: **22 persons.**

- **Example 2 — In a class of 40, Rohan is 13th from the top in merit. What is his rank from the bottom?**

- Step: Rank from bottom = 40 - 13 + 1. - Answer: **28th from the bottom.**

Circular Seating — The Method

People sit around a circle, usually facing the centre. Two facts control everything:

- When a person **faces the centre**, their **left side is the clockwise direction** and their right side is anti-clockwise. (Sit facing inward: your left hand points along the circle in the clockwise sense.)
- When a person **faces outward**, left and right reverse: left becomes anti-clockwise.

Method steps:

- Step 1: Draw a circle with as many marks as people. Fix **one person's** seat — in a circle, a starting point costs nothing.
- Step 2: Place people with **direct clues** first ("B sits next to A", "C is second to the left of D").
- Step 3: Use **negative clues** last ("X is not a neighbour of Y") to eliminate positions.
- Step 4: Remember **opposite seats**: in an even-numbered circle, the person exactly halfway round sits opposite.

Linear Seating — The Method

In a row, first fix **which way they face** — if all face north, left and right are the viewer's left and right; if they face south, left and right flip. Place the end-position clues first ("P sits at an end", "Q is second from the right end"), then neighbours, then gaps ("three persons sit between A and B" means a gap of three seats, so their positions differ by 4).

- **Example — Six persons A to F sit in a row facing north. B is between A and C. D sits at the right end. E is to the immediate left of D. Who sits at the left end if F is not next to A?**

- Step: D at right end (seat 6), E at seat 5. A-B-C or C-B-A must occupy three seats together from seats 1-4; F takes the remaining seat. If F is not next to A, place F at seat 1 only if A is not at seat 2 — try C-B-A at seats 2-4: then seat 1 = F, and F's neighbour is C, not A. Works. - Answer: **F sits at the left end (order: F, C, B, A, E, D).**

MEMORY HOOK: "Between" problems: seats between two people = difference of their positions minus 1.

EXAM TRAP: In circular seating, students use their own left-right. When persons face the centre, THEIR left is clockwise — the opposite of what you first imagine. And in ranking, forgetting the **-1** is the single most common error in the whole topic.

Shapes and Their Sub-Sections

These questions show a figure full of lines and ask: how many triangles? How many squares or rectangles? Brute counting misses figures hiding inside other figures. Use the **label-and-count method**: name every small region with a letter, then count figures in order of size — singles first, then figures made of 2 parts, then 3 parts, and so on — writing each one down. Written lists do not

repeat and do not miss.

Counting Triangles — The Shortcut

For the very common figure — a big triangle with lines drawn from the top vertex down to the base, dividing the base into **n** parts — the number of triangles is:

n(n + 1) / 2

So a base divided into 4 parts gives 4 x 5 / 2 = **10 triangles**. The same shortcut counts rectangles in a row/strip divided into n parts, and the number of handshakes-style pairings. When horizontal lines also cut the figure, count level by level and add.

Counting Squares and Rectangles

- In a square grid of **n x n** small squares, total squares = **$1^2 + 2^2 + ... + n^2$**. For 2x2: 4 + 1 = 5 squares. For 3x3: 9 + 4 + 1 = 14.
- In a rectangular grid of rows **m** and columns **n**, total rectangles = **(1 + 2 + ... + m) x (1 + 2 + ... + n)** = [m(m+1)/2] x [n(n+1)/2]. Squares are included in rectangles — unless the question says "rectangles but not squares".
- **Example — How many squares are in a 2 x 2 board?**

- Step: Small squares = 4. One big square made of all four = 1. - Answer: **4 + 1 = 5 squares.**

Embedded Figures

A small figure is hidden inside a big jumbled figure; you must pick which option's outline contains it exactly (same size, rotation allowed in most tests). Method: trace the small figure's **sharp corners and longest line** first and search for them in the big figure.

Mirror and Water Images

- **Mirror image**: left and right swap, top and bottom stay. The mirror is imagined standing **vertically** beside the figure. Letters like A, H, I, O, X look the same in a mirror.
- **Water image**: top and bottom swap, left and right stay — the figure flips as in still water below it. Letters like C, D, E, H, I, O, X look almost the same in water.
- In both, the part **nearest** the mirror/water line stays nearest after the flip.

Paper Folding and Cutting (in brief)

A paper is folded (shown by arrows) and a cut or punch is made; you choose the unfolded pattern. Method: work **backwards** — undo the last fold first, mirroring the cut at each step. Each fold doubles the holes: one punch through 2 folds gives 4 holes, through 3 folds gives 8, placed symmetrically across each fold line.

MEMORY HOOK: Count small to large, write every figure's name — the list is your proof.

EXAM TRAP: In stacked-triangle figures students count only the smallest triangles. The big outer triangle and the in-between sizes are triangles too — the n(n+1)/2 shortcut exists precisely because humans miss them.

Percentage and Its Family

Percentage Basics

Percentage means "out of 100": x percent = **x/100**. To change a fraction into a percentage, multiply by 100; to change a percentage into a fraction, divide by 100. Percentage questions become fast when you know the standard fraction equivalents by heart.

Fraction	1/2	1/3	1/4	1/5	1/6	1/7	1/8
Percent	50%	33.33%	25%	20%	16.67%	14.28%	12.5%

Fraction	1/9	1/10	1/11	1/12	3/4	2/5	5/8
Percent	11.11%	10%	9.09%	8.33%	75%	40%	62.5%

Key tools:

- **x% of y = y% of x**. So 48% of 50 = 50% of 48 = 24. Use whichever side is easier.
- **Percentage change = (change / original value) x 100**. The original value always goes below the line.
- **Successive change**: if a value rises by a% then by b%, the net change is **a + b + (a x b)/100** percent. For a fall, take that percentage as negative. Two changes of +10% and -10% give 10 - 10 - 100/100 = **-1%** — a net loss, never zero.
- **Example 1 — A town's population of 20,000 rises 10% in one year and 10% again the next. New population?**

- Step: Net rise = 10 + 10 + (10x10)/100 = 21%. New population = 20,000 x 121/100. - Answer: **24,200.**

- **Example 2 — The price of sugar rises by 25%. By what percent must a family cut consumption so that spending stays the same?**

- Step: Cut needed = rise/(100 + rise) x 100 = 25/125 x 100 = 20%. - Answer: **Cut consumption by 20%.** (Reverse rule: if price falls by r%, consumption can rise by r/(100 - r) x 100 percent.)

> **MEMORY HOOK:** When price rises by r%, consumption cut = **r/(100 + r)** of the new amount — the denominator is the NEW price level, never 100.

Ratio, Proportion and Partnership

A **ratio** a : b compares two quantities of the **same kind and unit**; as a fraction it is a/b. Ratios can be multiplied or divided by the same number freely.

- **Combining ratios**: if A : B = 2 : 3 and B : C = 4 : 5, make B equal (LCM 12): A : B = 8 : 12 and B : C = 12 : 15, so **A : B : C = 8 : 12 : 15**.
- **Proportion**: a : b = c : d means **a x d = b x c** (product of extremes = product of means). In one line: **componendo-dividendo** says if a/b = c/d, then (a + b)/(a - b) = (c + d)/(c - d) — handy when a question gives (x + y)/(x - y) directly.
- **Partnership rule**: when partners invest for different times, profit is shared in the ratio of **capital x time** for each partner. Equal time means simple capital ratio; equal capital means simple time ratio.
- **Example 1 — Divide Rs. 1,100 among three people in the ratio 3 : 5 : 3.**

- Step: Total parts = 3 + 5 + 3 = 11. Each part = 1100 / 11 = Rs. 100. Shares = 3 x 100, 5 x 100, 3 x 100. - Answer: **Rs. 300, Rs. 500, Rs. 300.**

- **Example 2 — A invests Rs. 5,000 for 12 months; B invests Rs. 8,000 for 5 months. Total profit Rs. 5,000. Find A's share.**

- Step: Capital x time: A = 5000 x 12 = 60,000; B = 8000 x 5 = 40,000. Ratio = 3 : 2. A's share = 3/5 of 5,000. - Answer: **A gets Rs. 3,000, B gets Rs. 2,000.**

Profit and Loss

Term	Formula
Profit	SP - CP (when SP > CP)
Loss	CP - SP (when CP > SP)
Profit %	(Profit / CP) x 100
Loss %	(Loss / CP) x 100
SP (at profit)	CP x (100 + profit%) / 100
SP (at loss)	CP x (100 - loss%) / 100
Discount	Marked Price - Selling Price
Discount %	(Discount / Marked Price) x 100

Profit and loss percentages are **always calculated on Cost Price**; discount is **always on Marked Price**. This one distinction settles half the chapter. **Successive discounts** of a% and b% equal a single discount of **a + b - (a x b)/100** percent.

- **Example 1 — A cycle bought for Rs. 2,000 is sold for Rs. 2,300. Profit percent?**

- Step: Profit = 300. Profit% = 300/2000 x 100. - Answer: **15% profit.**

- **Example 2 — A jacket marked Rs. 1,000 gets two discounts of 20% and 10%. Final price?**

- Step: Single equal discount = 20 + 10 - (20x10)/100 = 28%. Price = 1000 x 72/100. - Answer: **Rs. 720.** (Not Rs. 700 — successive discounts never simply add.)

EXAM TRAP: The deadliest trap of this chapter — profit percent on **selling price**. Always divide by CP. Second trap: adding successive discounts (20% + 10% = 30%) instead of using the net formula (28%).

PYQ PATTERN: RAS repeats the consumption-cut question and successive-discount question almost every cycle — both shortcuts above are direct marks.

Simple and Compound Interest

Interest is the rent paid for using money. **Principal (P)** is the borrowed sum, **Rate (R)** is percent per year, **Time (T)** is in years.

Simple Interest (SI)

Under simple interest, interest is earned **only on the original principal** every year.

- **SI = (P x R x T) / 100**
- **Amount = P + SI = P(1 + RT/100)**
- If a sum **doubles** at SI, the interest equals P, so RT = 100. At 5% a sum doubles in 20 years by SI.

Compound Interest (CI)

Under compound interest, each year's interest **joins the principal** and itself earns interest.

- **Amount = P(1 + R/100)^T**
- **CI = Amount - P**
- **CI - SI for 2 years = P(R/100)²**
- **CI - SI for 3 years = P(R/100)² x (3 + R/100)**
- Compounding **half-yearly**: rate becomes R/2, time becomes 2T. **Quarterly**: rate R/4, time 4T. More frequent compounding always gives a little more interest.
- **Example 1 — Find SI on Rs. 6,000 at 8% per year for 3 years.**

- Step: SI = 6000 x 8 x 3 / 100 = 144,000 / 100 = 1,440. - Answer: **SI = Rs. 1,440; Amount = Rs. 7,440.**

- **Example 2 — Find CI on Rs. 10,000 at 10% for 2 years, and the CI-SI difference.**

- Step: Amount = 10,000 x (1.1)² = 10,000 x 1.21 = 12,100, so CI = 2,100. SI = 10,000 x 10 x 2/100 = 2,000. Check by formula: difference = P(R/100)² = 10,000 x 0.01 = 100. - Answer: **CI = Rs. 2,100; CI - SI = Rs. 100.**

MEMORY HOOK: CI minus SI for 2 years is just "interest on the first year's interest" = P(R/100)² — if you remember the story, you never forget the formula.

EXAM TRAP: In CI problems with half-yearly compounding, students halve the rate but forget to double the time (or the reverse). Both changes must happen together.

Area and Perimeter of Plane Figures

Perimeter is the length of the boundary (one-dimensional, in cm or m). **Area** is the surface covered (two-dimensional, in cm² or m²). Most errors in this chapter are unit errors or radius-diameter mix-ups, not formula errors — learn the table and then guard the units.

Formula Table

Figure	Perimeter	Area
Square (side a)	4a	a²
Rectangle (l x b)	2(l + b)	l x b
Triangle (base b, height h)	a + b + c	(1/2) x b x h
Triangle by Heron	a + b + c	root of [s(s-a)(s-b)(s-c)], s = (a+b+c)/2
Equilateral triangle (side a)	3a	(root 3 / 4) x a²
Parallelogram (base b, height h)	2(a + b)	b x h
Rhombus (diagonals d1, d2)	4a	(1/2) x d1 x d2
Trapezium (parallel sides a, b; gap h)	sum of all sides	(1/2) x (a + b) x h
Circle (radius r)	2 x pi x r (circumference)	pi x r²

Use **pi = 22/7** when the radius or diameter is a multiple of 7, and **pi = 3.14** otherwise — the question's numbers tell you which.

Units and Conversions

- **1 m = 100 cm; 1 m² = 10,000 cm²** (areas square the conversion — the most tested point).
- **1 hectare = 10,000 m²; 1 km = 1,000 m; 1 km² = 1,000,000 m².**
- A path of width w around a rectangle adds **2w** to both length and breadth for the outer figure.
- **Example 1 — Find the area and perimeter of a rectangle 12 m long and 8 m wide.**

- Step: Area = 12 x 8 = 96. Perimeter = 2(12 + 8) = 40. - Answer: **Area 96 m², perimeter 40 m.**

- **Example 2 — A circular park has radius 14 m. Find its area and circumference (pi = 22/7).**

- Step: Area = (22/7) x 14 x 14 = 22 x 2 x 14 = 616. Circumference = 2 x (22/7) x 14 = 88. - Answer: **Area 616 m², circumference 88 m.**

MEMORY HOOK: For circles, if the radius is 7, 14, 21... the 7 cancels — the paper-setter has already chosen 22/7 for you.

EXAM TRAP: Using diameter where the formula needs radius (area uses r^2 — putting d in place of r makes the answer **four times** too big). Also, writing area in plain metres instead of square metres loses the mark in descriptive checking and signals a guess.

Data Interpretation

Data Interpretation (DI) gives a **table, bar diagram, line graph or pie chart** and asks 4-5 questions on it. The maths is only percentage, average and ratio — the real skill is reading the right number quickly and not calculating more than the question needs.

How to Read Each Type

- **Table**: note the units in the heading (thousands? lakhs? tonnes?) before touching any number. Many wrong answers are right numbers in wrong units.
- **Bar diagram**: compare heights; for "how much more/less percent", always divide by the **first-named (base)** value.
- **Line graph**: the slope shows growth; a steeper line means faster growth even when the total is smaller.
- **Pie chart**: the whole circle is **360 degrees = 100%**. **Degree of a share = share% x 3.6**, or **share = (angle / 360) x total**. Conversely, percent = angle / 3.6.

The Three Question Types Asked

Type	What it asks	Tool
Value	"How much / how many in year X?"	Direct reading, or angle-to-value in a pie
Comparison / ratio	"Ratio of A to B?" or "A is what % of B?"	Divide, simplify, x100 for percent
Growth %	"Percent increase from year X to Y?"	(New - Old) / Old x 100

Speed methods:

- **Estimate first**: in multiple choice, round the numbers (4,980 to 5,000) and eliminate wild options before exact calculation.

- **Average shortcut**: average = total / count; if only the average is asked and numbers are close, work with **deviations from a guessed middle value**.
- **Example 1 (Pie) — A family's monthly income is Rs. 36,000. In the pie chart, food takes an angle of 120 degrees. How much is spent on food?**

- Step: Food share = 120/360 = 1/3 of income. Amount = 36,000 x 1/3. - Answer: **Rs. 12,000.**

- **Example 2 (Table) — A school's students: 2019 = 400, 2020 = 500, 2021 = 450, 2022 = 600, 2023 = 550. Find the average per year and the growth percent from 2019 to 2022.**

- Step: Total = 400 + 500 + 450 + 600 + 550 = 2,500; average = 2,500/5 = 500. Growth 2019 to 2022 = (600 - 400)/400 x 100 = 50%. - Answer: **Average 500 students; growth 50%.**

EXAM TRAP: In growth questions students divide by the NEW value. Growth percent is always measured against the OLD (earlier) value. And in pie charts, never confuse degrees with percent — 120 degrees is 33.33%, not 120%.

Mean, Median, Mode, Permutation, Combination and Probability

Mean — Three Kinds

- **Arithmetic Mean (AM)** = (sum of observations) / (number of observations).
- **Geometric Mean (GM)** of two numbers a and b = **root of (a x b)**. For n numbers, GM is the nth root of their product.
- **Harmonic Mean (HM)** of two numbers a and b = **2ab / (a + b)**.
- Relation: **AM >= GM >= HM**, with equality only when all numbers are equal. For two numbers: **GM^2 = AM x HM** — given any two of the three means, the third follows at once.
- Classic HM use: **average speed** for equal distances at speeds x and y is **2xy/(x + y)** — the harmonic mean, never the simple average.

Median and Mode

- **Median** is the middle value of arranged data. For **n odd**, the ((n+1)/2)th value. For **n even**, the average of the (n/2)th and (n/2 + 1)th values.
- For simple **grouped data**, median = **L + [(N/2 - cf) / f] x h**, where L is the lower limit of the median class, cf the cumulative frequency before it, f its frequency, h the class width.
- **Mode** is the most frequent value — the value with the highest frequency.
- **Empirical relation**: **Mode = 3 x Median - 2 x Mean**. Given any two of mean, median, mode, the third comes out of this line.
- **Example — Find mean, median and mode of 4, 8, 6, 8, 10.**

- Step: Mean = (4 + 8 + 6 + 8 + 10)/5 = 36/5 = 7.2. Arranged: 4, 6, 8, 8, 10 — middle value = 8. Mode = 8 (appears twice). - Answer: **Mean 7.2, Median 8, Mode 8.** Note: the relation Mode = 3 Median - 2 Mean is empirical (approximate) — it holds well for smooth, large data sets. When an exam question gives you two of the three values and asks for the third, apply it directly; do not expect it to match exactly on a tiny raw list like this one.

MEMORY HOOK: Mean uses every value, median uses only position, mode uses only frequency — the question's wording tells you which one is safe against extreme values (that is always the median).

Permutation and Combination

- **Permutation (nPr)** counts **arrangements** — order matters. **nPr = n! / (n - r)!**
- **Combination (nCr)** counts **selections** — order does not matter. **nCr = n! / [r! x (n - r)!]**
- **n! (factorial)** = n x (n-1) x ... x 1; remember 0! = 1 and nCr = nC(n-r).
- Quick test: "arrange, rank, seat, president-secretary" = permutation; "select, choose, team, committee" = combination.
- **Example — In how many ways can a team of 3 be chosen from 8 players? And in how many ways can 3 prizes (1st, 2nd, 3rd) be given to 8 players?**

- Step: Team = selection: 8C3 = 8! / (3! x 5!) = (8 x 7 x 6)/(3 x 2 x 1) = 56. Prizes = arrangement: 8P3 = 8 x 7 x 6 = 336. - Answer: **56 ways to select; 336 ways to award prizes.**

Probability (Simple Problems)

Probability = (number of favourable outcomes) / (total number of equally likely outcomes). Probability always lies between **0 (impossible) and 1 (certain)**.

Play tool	Facts to remember
Coin	1 coin: 2 outcomes; 2 coins: 4 outcomes (HH, HT, TH, TT)
Dice	One die: 6 outcomes; two dice: 36 outcomes
Cards	52 cards: 4 suits (spades, hearts, diamonds, clubs) of 13 each
Cards — colours	26 red (hearts + diamonds), 26 black (spades + clubs)
Cards — face cards	12 face cards (Jack, Queen, King of each suit); 4 aces; 16 "honour" cards (A, J, Q, K) in some books — use face = 12 unless stated

- **Example — A card is drawn from a well-shuffled pack of 52. Find the probability that it is a king, and that it is red.**

- Step: Kings = 4, so P(king) = 4/52 = 1/13. Red cards = 26, so P(red) = 26/52 = 1/2. - Answer: **P(king) = 1/13; P(red) = 1/2.**

EXAM TRAP: Two classic traps close this book. First, average speed: equal distances at 40 and 60 km/h give **48 km/h** (HM), not 50 — the simple mean is the planted wrong option. Second, in probability, "at least one head in two coins" is 3/4, not 1/2 — count TT as the only failure. And in combinations, never use nPr when the question only asks to **choose**.

PYQ PATTERN: RAS probability questions stay simple — one die, two coins, one card. Master the deck facts table and the 36-outcome two-dice grid and these marks are free.